

INDU 211 · Comprehensive Topic Guide (Part 7)

Chapter 8: Quality Control & Statistical Process Control

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Built on the teacher's Lecture 11 (Chapter 8, Quality Control), with depth from the course textbook (Hicks, Chapter 8).

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1. What Quality Means

The slides open with eight definitions (Lecture 11, slides 3–10). Learn who said what: exams test the match.

Source	Definition of quality
ISO	The totality of features and characteristics that bear on the ability to satisfy stated or implied needs
Gitlow et al.	The extent to which customers believe the product surpasses their needs and expectations
Feigenbaum	The total composite of marketing, engineering, manufacture and maintenance characteristics through which the product meets customer expectations
Imai	Anything which can be improved
Crosby	Conformance to requirements
Shewhart	Conformance to specified standards
Juran	Fitness for use
Deming	A predictable degree of uniformity and dependability at low cost, suited to the market

Two viewpoints: the **manufacturing-based** approach (supply side: engineering and manufacturing practice, conformance) and the **value-based** approach (quality in terms of cost and price: performance at an acceptable price).

Dimensions of quality for goods (slide 14): performance, reliability and durability, conformance, serviceability, appearance, features, safety. **Service attributes (slide 15):** reliability, communication, responsiveness, competence, courtesy, access.

Determinants of quality (slide 22): quality of design, capability of the production process, quality of conformance, quality of customer service.

From the textbook (Hicks, §8.4): Deming's 14 points (first eight, paraphrased). (1) Create constancy of purpose toward improvement. (2) Adopt the new philosophy. (3) **Cease dependence on inspection:** build quality in from the start. (4) Stop awarding business on price alone; minimise total cost, with long-term single suppliers. (5) Improve the system of production and service constantly and forever. (6) Institute training on the job. (7) Institute leadership. (8) Drive out fear. Point 3 is the idea behind moving from quality *control* (detection) to quality *assurance* (prevention).

2. Quality Management, Standards and Assurance

ISO quality management system standards (slide 25): they specify **what** is required, not how to do it; certification is done by a third party and renewed every **three years**.

Standard	Covers
ISO 9001	Quality management (products and services meet customer needs)
ISO 14001	Environmental management
ISO 45001	Health and safety at work
ISO 27001	Information security

	Quality control (QC)	Quality assurance (QA)
Focus	Detection of defects	Prevention of defects
Goal	Identify and fix issues in the output	Make the process effective and consistent
Activities	Inspection, testing, sampling, rework	Standard procedures, process audits, training, documentation, certification

3. The Costs of Quality

Category	When the money is spent	Examples
Prevention	Before defects occur	Quality planning, training, designing quality systems, reporting
Appraisal	Evaluating products	Testing, inspection, quality audits
Internal failure	Defect found before delivery	Scrap, rework, retest, downtime, yield loss, disposition
External failure	Defect found after delivery	Complaints, returns, warranty, allowances, loss of goodwill

Total quality cost = the difference between the actual cost and what it would be with no possibility of defects (slide 33).

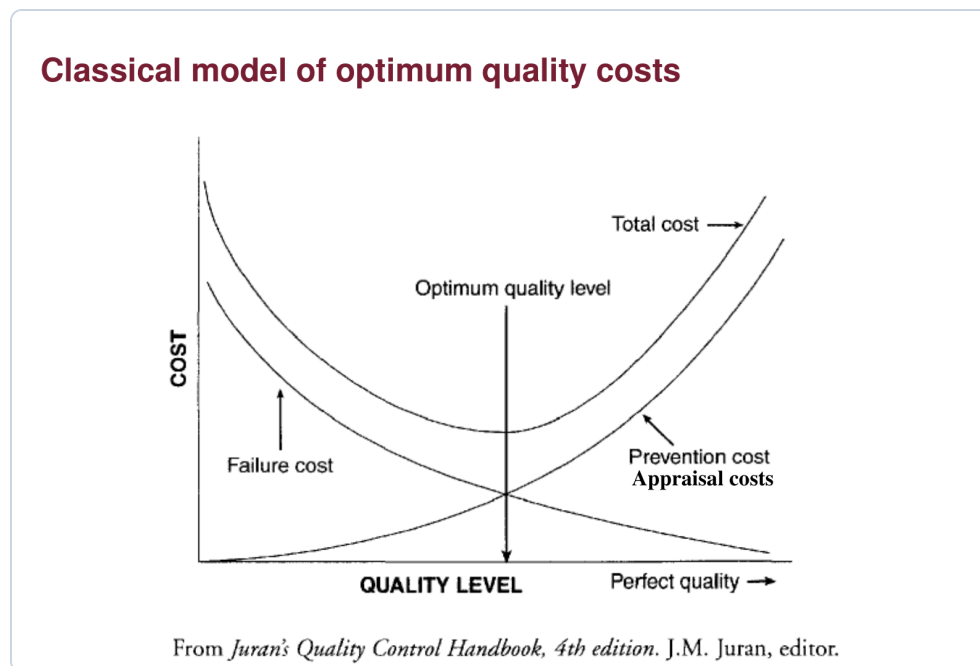
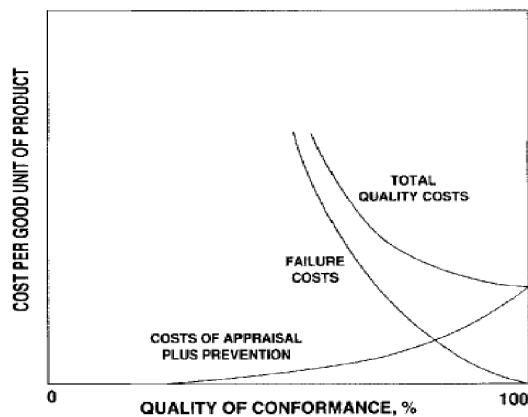


Figure 1: Classical model of optimum quality costs, Lecture 11, slide 34 (Juran's Quality Control Handbook).

Reading the graph: as quality level rises toward perfect, **failure costs fall** but **prevention and appraisal costs rise** steeply. Their sum (total cost) is U-shaped, with a minimum at an "optimum quality level" **below** perfection. The classical view says perfection is not worth paying for.

New model of optimum quality costs



From *Juran's Quality Control Handbook, 4th edition*. J.M. Juran, editor.
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- Many organizations have demonstrated that with good design, processes and automation, you can drive defects to extremely low levels at manageable cost.

Figure 2: New model of optimum quality costs, Lecture 11, slide 35.

Reading the graph: in the new model, prevention and appraisal costs **do not explode** near 100% conformance, because good design, capable processes and automation drive defects down cheaply. Total cost keeps falling all the way to 100% conformance, so the optimum is **zero defects**. This is the economic argument behind six sigma.

4. Statistical Process Control: Natural vs Assignable Variation

SPC is a data-driven method for distinguishing **natural (random)** variation from **assignable** variation (slide 37).

Random (natural, common-cause)	Assignable (special-cause)
Many tiny, inherent causes	An identifiable source with significant impact
Small fluctuations in raw material; gradual tool wear; temperature, humidity; operator reaction time	Equipment failure; lost calibration; changeover mistake; new supplier; operator error; power surge
Leave the process alone	Find the cause and remove it

Overview of the Quality Control Process

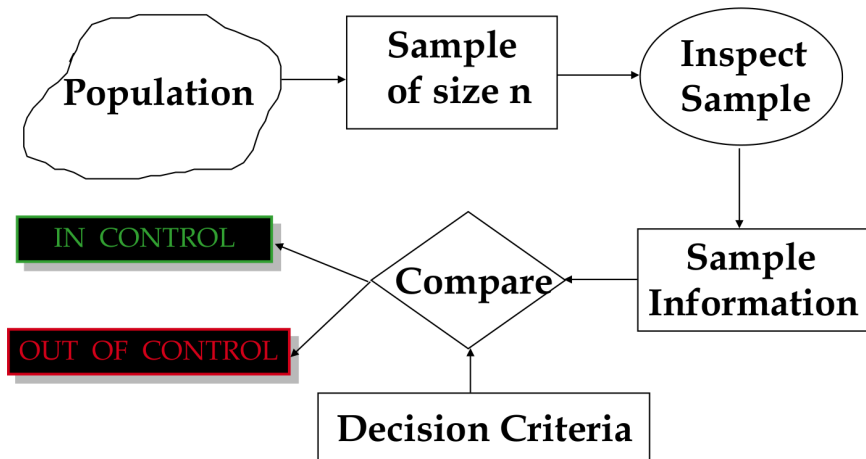


Figure 3: Overview of the quality control process, Lecture 11, slide 38.

Reading the loop: take a **sample of size n** from the population, **inspect** it, turn it into **sample information** (mean, range, fraction defective), **compare** it with the **decision criteria** (control limits), and conclude **in control** or **out of control**.

Control Chart

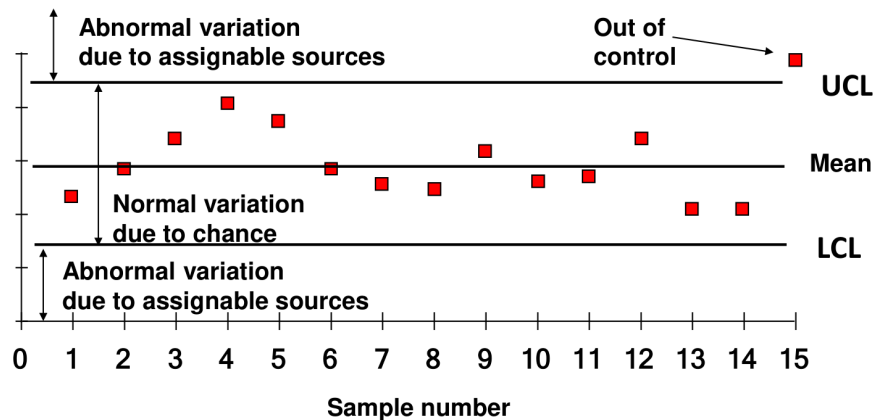


Figure 4: Control chart, Lecture 11, slide 39.

Reading the chart: points between the **UCL** and **LCL** scattered around the mean are normal variation due to chance. A point above the UCL or below the LCL signals **abnormal variation due to an assignable source**.

Tracking Improvements

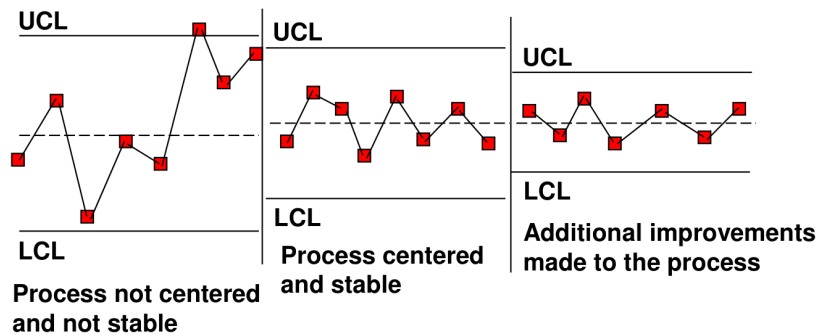


Figure 5: Tracking improvements, Lecture 11, slide 40.

Reading the three panels: (left) the process is not centred and not stable, with points outside the limits; (middle) after removing assignable causes it is centred and stable; (right) further improvements **narrow the limits**, meaning less natural variation. Control charts both detect problems and document improvement.

From the textbook (Hicks, §8.10): the seven basic SPC tools. Flowchart, cause-and-effect (fishbone) diagram, data collection (check) sheet, Pareto analysis, histogram, scatter plot, and designed experiments. The control chart is the monitoring tool that ties them together.

5. The Normal Distribution Refresher

Control limits rest on the normal distribution: symmetric and bell-shaped, with mean = median = mode at the centre.

- Variance: $\sigma^2 = \frac{\sum (X - \mu)^2}{n}$. For 1, 2, 3: mean 2, variance $\frac{1+0+1}{3} = 0.667$.
- Standardising: $Z = \frac{X - \mu}{\sigma}$ is the number of standard deviations from the mean. A score of 70 with mean 50 and $\sigma = 10$ is $Z = 2$.

Review of Normal Distribution

The odds of a particular score, or set of scores, falling within a particular region are equal to the percentage of the total area occupied by that region

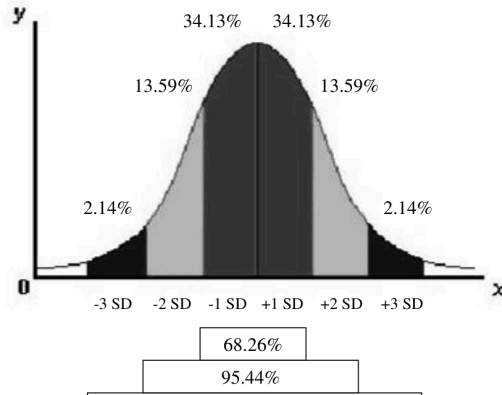


Figure 6: Review of the normal distribution, Lecture 11, slide 45.

Range	Area inside
$\mu \pm 1\sigma$	68.26%
$\mu \pm 2\sigma$	95.44%
$\mu \pm 3\sigma$	99.72%

So **3 σ control limits** contain 99.72% of points from a stable process: a point outside is very unlikely to be chance (about 3 in 1,000).

6. Control Charts for Variables: X-bar and R

For **measured** (continuous) data, two charts are kept together: the $\bar{X}$ **chart** monitors central tendency, and the R **chart** monitors variability.

$$\bar{X} \text{ chart: } \bar{\bar{X}} \pm A_2 \bar{R}, \quad R \text{ chart: } LCL = D_3 \bar{R}, \quad UCL = D_4 \bar{R}, \quad \hat{\sigma} = \frac{\bar{R}}{d_2}$$

where A_2 , D_3 , D_4 , d_2 depend on the sample size n .

SPC charts coefficients

Table 8.1. CONTROL CHART FACTORS

Subgroup Size, n	LCL_R Factor, D_3	UCL_R Factor, D_4	$CL_{\bar{X}}$ Factor, A_2	$\bar{R}/\sigma$ Ratio, d_2
2	0	3.267	1.880	1.128
3	0	2.575	1.023	1.693
4	0	2.282	0.729	2.059
5	0	2.115	0.577	2.326
6	0	2.004	0.483	2.534
7	0.076	1.924	0.419	2.704
8	0.136	1.864	0.373	2.847
9	0.184	1.816	0.337	2.970
10	0.223	1.777	0.308	3.078
11	0.256	1.744	0.285	3.173
12	0.284	1.716	0.266	3.258
13	0.308	1.692	0.249	3.336
14	0.329	1.671	0.235	3.407
15	0.348	1.652	0.223	3.472
16	0.364	1.636	0.212	3.532

Figure 7: SPC chart coefficients, Lecture 11, slide 50.

Reading the table: find the row for your sample size n , then read A_2 (for the $\bar{X}$ limits), D_3 and D_4 (for the R limits) and d_2 (to estimate σ). For $n = 4$: $A_2 = 0.729$, $D_3 = 0$, $D_4 = 2.282$, $d_2 = 2.059$.

Lecture example: soft-drink fill weights (slides 56–57)

$n = 20$, $\bar{X} = 16$ oz, $\bar{R} = 0.4$ oz, $A_2 = 0.18$, $D_3 = 0.41$, $D_4 = 1.59$.

$$UCL_{\bar{X}} = 16 + 0.18(0.4) = 16.07, \quad LCL_{\bar{X}} = 16 - 0.18(0.4) = 15.93$$

$$UCL_R = 1.59(0.4) = 0.636, \quad LCL_R = 0.41(0.4) = 0.164$$

Solution – sample means

- $n = 20$
- $A_2 = 0.18$
- $D_3 = 0.41$
- $D_4 = 1.59$

$\bar{R}$

$= 0.4$

$\bar{\bar{X}} = 16$

- $LCL_R = D_3\bar{R} = 0.41(0.4) = 0.164$ ounces
- $UCL_R = D_4\bar{R} = 1.59(0.4) = 0.636$ ounces
- Upper limit : $\bar{\bar{X}} + A_2\bar{R} = 16 + 0.18(0.4) = 16.07$ ounces
- Lower limit : $\bar{\bar{X}} - A_2\bar{R} = 16 - 0.18(0.4) = 15.93$ ounces

Sample	X-bar	R
1	16.05	0.20
2	16.04	0.25
3	15.98	0.62
4	15.91	0.71
5	16.02	0.58
6	16.09	0.37
7	15.95	0.35
8	16.06	0.21
9	15.94	0.29
10	15.97	0.46

Out of control

Figure 8: Solution: sample means, Lecture 11, slide 57.

Reading the result: sample 4 ($\bar{X} = 15.91$, below 15.93, and $R = 0.71$, above 0.636) and sample 6 ($\bar{X} = 16.09$, above 16.07) are **out of control**; the filling process needs investigation.

Lecture example: loan processing times (slide 58)

Five samples of $n = 4$: $\bar{\bar{X}} = 12.11$ min, $\bar{R} \approx 0.05$ (exactly 0.046), $A_2 = 0.729$, $D_4 = 2.282$:

$$\bar{X} \text{ limits} = 12.11 \pm 0.729(0.05) = 12.07 \text{ to } 12.14, \quad UCL_R = 2.282(0.05) = 0.11, \quad LCL_R = 0$$

7. Control Charts for Attributes: the p Chart

For **pass/fail** (discrete) data, the p chart tracks the fraction defective:

$$\bar{p} \pm 3\sqrt{\frac{\bar{p}(1 - \bar{p})}{n}}$$

Lecture example: bolts (slides 60–62)

Expected 4% defective, samples of $n = 100$:

$$\sigma_p = \sqrt{\frac{0.04(0.96)}{100}} = 0.0196, \quad UCL = 4 + 3(1.96) = 9.88\%, \quad LCL = 4 - 5.88 = -1.88\% \rightarrow$$

Solution

$$\bar{p} \pm 3\sqrt{\bar{p}(1-\bar{p})/n}$$

- Upper limit: $4+3(1.9596) = 9.88 \%$ **Out-of-control process**
- Lower limit: $4-3(1.9596) = -1.88$ or 0

Sample Number	Percent Defectives
1	4
2	3
3	3
4	6
5	1
6	9
7	5
8	12

Figure 9: p chart solution, Lecture 11, slide 62.

Reading the result: day 8 (12% defective) is above 9.88%, so the process is **out of control** on that day. The negative lower limit is replaced by 0 because a fraction defective cannot be negative.

8. Process Capability and Six Sigma

Control charts ask "is the process **stable**?". Capability asks "is a stable process **good enough** for the specifications?".

$$C_p = \frac{USL - LSL}{6\hat{\sigma}}$$

Process Capability

- Indicates how capable a process is in meeting customer requirements
- $C_p = \frac{USL - LSL}{6\hat{\sigma}}$
- USL: upper specification limit
- LSL: lower specification limit
- $\hat{\sigma}$: Process standard deviation

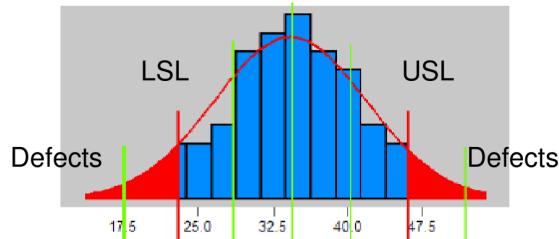


Figure 10: Process capability, Lecture 11, slide 65.

Reading the histogram: the bars are the process output; LSL and USL are the specification limits set by the designer. Output beyond them (red tails) is defective. C_p compares the width allowed ($USL - LSL$) with the width the process actually uses (6σ): $C_p > 1$ means the natural spread fits inside the specification.

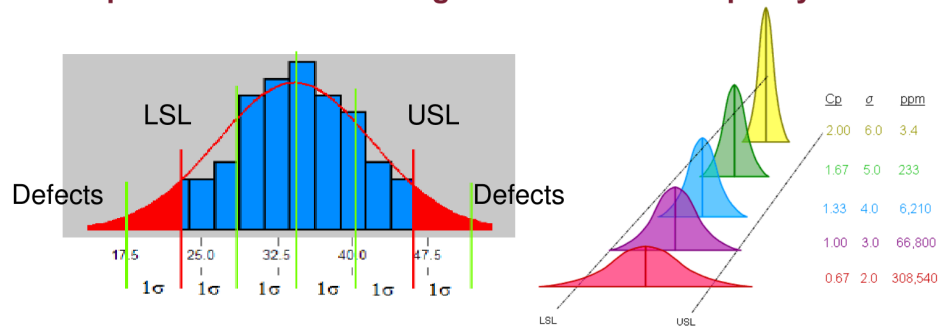
Worked example (Fall 2020 final, Best-Wood beams): specs 24 ± 2.2 cm; after removing two out-of-control samples, $\bar{R} = 1.392$ with $n = 4$ ($d_2 = 2.059$):

$$\hat{\sigma} = \frac{1.392}{2.059} = 0.676, \quad C_p = \frac{4.4}{6(0.676)} = 1.08, \quad k = \frac{2.2}{0.676} = 3.25\sigma$$

Six sigma

- Aim: virtually error-free performance, **3.4 defects per million** opportunities (slide 66).
- Sigma level: $k\sigma = \frac{\text{Tolerance}}{2}$, the distance from target to a specification limit in standard deviations.
- Improve the **mean** (production adjustment: re-centre) or the **variance** (system change: reduce spread).

Comparison between Six Sigma and traditional quality models



Tolerance=USL-LSL

- A way to measure the sigma level of a process is by calculating k in the formula below:
 - The distance from the target to the upper or lower specification (half the tolerance), measured in terms of standard deviation of the inherent variation

Figure 11: Six sigma vs traditional quality models, Lecture 11, slide 68.

Reading the figure: the same tolerance ($USL - LSL$) is shown with a wide traditional distribution (3σ fits, so the tails are defective) and a narrow six-sigma distribution (6σ fits inside each half-tolerance, so almost nothing falls outside). The curves on the right show what shifting the mean does: the defect rate rises sharply as the process drifts off-centre.

Six Sigma methodology

- Quality improvement via the DMAIC model
- A structured problem-solving process provides all employees with a common language and a set of tools to communicate with each other

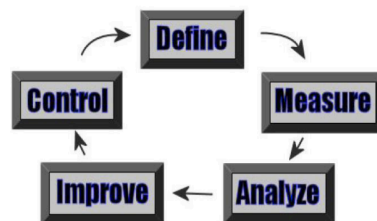


Figure 12: Six sigma methodology, Lecture 11, slide 70.

DMAIC: Define → Measure → Analyze → Improve → Control, a structured problem-solving cycle that gives everyone a common language.

9. Exam Checklist

- [] Match each guru to his definition (Juran: fitness for use; Crosby: conformance to requirements...).
- [] QC = detection vs QA = prevention.
- [] Classify costs: prevention, appraisal, internal failure (before delivery), external failure (after).
- [] Classical (optimum below 100%) vs new (optimum at zero defects) cost models.
- [] Random vs assignable causes; what a point outside the limits means.
- [] $\bar{X}$ limits $\bar{X} \pm A_2 \bar{R}$; R limits $D_3 \bar{R}, D_4 \bar{R}$; $\hat{\sigma} = \bar{R}/d_2$.
- [] p-chart limits; truncate a negative LCL at 0.
- [] $C_p = (USL - LSL)/6\hat{\sigma}$; six sigma = 3.4 defects per million; DMAIC.